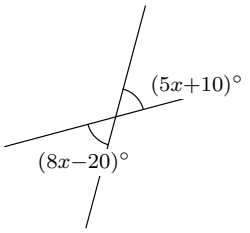
Geometry Review: Unit 2 — Angles & Parallel Lines

Station 1 of 4: Angle Pairs: Vertical, Linear, Complementary, Supplementary

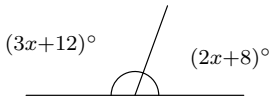
Name: _____ Date: _____

Set up and solve an equation for each problem. Include degree symbols in your answers. Diagrams are not drawn to scale.

1. The two labeled angles are vertical angles. Find x and the measure of each angle.



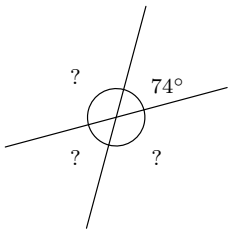
2. The two labeled angles form a linear pair. Find both angles.



3. Find the complement of $47°$.

4. Find the supplement of $128°$.

5. Two lines intersect as shown. Find the other three angles.

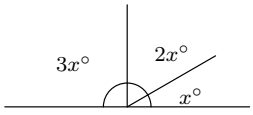


6. Two adjacent complementary angles measure $x°$ and $(x + 20)°$. Find both.

7. A linear pair measures $(4x - 5)°$ and $(2x + 11)°$. Find both angles.

8. Vertical angles measure $(2x + 30)°$ and $(4x - 10)°$. Find x and the angle measure.

9. The three adjacent angles form a straight angle. Find each angle.



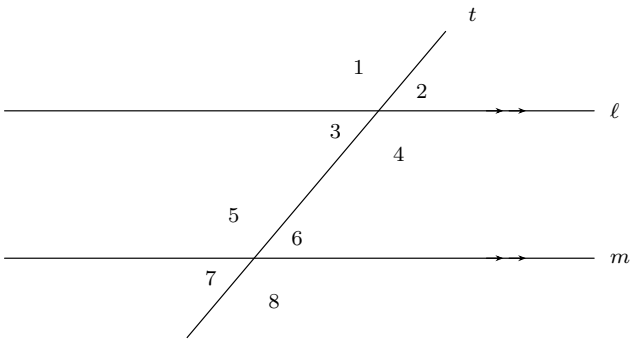
10. Two lines are perpendicular and form an angle of $(6x - 6)°$. Find x.

Geometry Review: Unit 2 — Angles & Parallel Lines

Station 2 of 4: Parallel Lines Cut by a Transversal

Name: _____ Date: _____

In the figure, $\ell \parallel m$ and t is a transversal. Use the angle relationships to find the missing measures. Diagrams are not drawn to scale.



1. If $m\angle 1 = 65°$, find $m\angle 5$.

2. If $m\angle 3 = 112°$, find $m\angle 6$.

3. If $m\angle 5 = 48°$, find $m\angle 3$.

4. If $m\angle 7 = 125°$, find $m\angle 2$.

5. $m\angle 2 = (4x + 7)°$ and $m\angle 6 = (6x - 13)°$. Find x and $m\angle 2$.

6. $m\angle 4 = (3x + 20)°$ and $m\angle 5 = (5x - 10)°$. Find x and $m\angle 4$.

7. $m\angle 3 = (2x + 15)°$ and $m\angle 5 = (3x + 10)°$. Find x and both angles.

8. $m\angle 1 = (7x - 4)°$ and $m\angle 8 = (5x + 14)°$. Find x and $m\angle 1$.

9. If $m\angle 1 = 70°$, find the measure of every other angle in the figure.

10. Complete: when parallel lines are cut by a transversal, alternate interior angles are _____ and same-side interior angles are _____.

Geometry Review: Unit 2 — Angles & Parallel Lines

Station 3 of 4: Proving Lines Are Parallel

Name: _____ Date: _____

Use the converse of the parallel line theorems. Show your work.

1. Corresponding angles measure $(5x + 4)^\circ$ and 64°. Find x so that the lines are parallel.

2. Alternate interior angles measure $(3x - 7)^\circ$ and $(2x + 9)^\circ$. Find x so that the lines are parallel, then the angle measure.

3. Same-side interior angles measure $(4x + 10)^\circ$ and $(6x + 30)^\circ$. Find x so that the lines are parallel, then both angles.

4. Alternate exterior angles measure $(8x - 12)^\circ$ and $(6x + 10)^\circ$. Find x so that the lines are parallel, then the angle measure.

5. Same-side interior angles measure $(x + 50)^\circ$ and $(2x + 10)^\circ$. Find x so that the lines are parallel.

6. Given $\angle 3 \cong \angle 6$ as alternate interior angles, which theorem proves the lines are parallel?

7. In a plane, two lines are each perpendicular to the same line. What can you conclude about the two lines?

8. Same-side interior angles measure 118° and 62°. Are the lines parallel? Explain.

9. Corresponding angles measure 75° and 85°. Are the lines parallel? Explain.

10. Line p has slope $\frac{2}{3}$. Line q passes through $(0, 1)$ and $(6, 5)$. Are p and q parallel?

Geometry Review: Unit 2 — Angles & Parallel Lines

Station 4 of 4: Triangle and Polygon Angle Sums

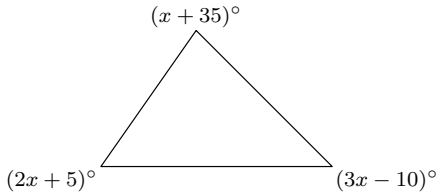
Name: _____ Date: _____

Use the Triangle Sum Theorem, the Exterior Angle Theorem, and polygon angle formulas. Diagrams are not drawn to scale.

1. Two angles of a triangle measure $52°$ and $67°$. Find the third angle.

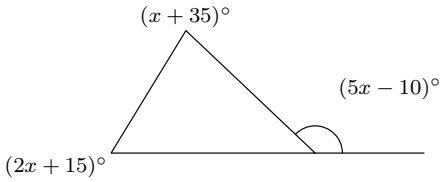
2. The angles of a triangle measure $x°$, $2x°$, and $3x°$. Find each angle.

3. The angles of a triangle measure $(2x + 5)°$, $(3x - 10)°$, and $(x + 35)°$. Find each angle.



4. The two remote interior angles of a triangle measure $48°$ and $71°$. Find the exterior angle.

5. An exterior angle of a triangle measures $(5x - 10)°$. The remote interior angles measure $(2x+15)°$ and $(x+35)°$. Find x and the exterior angle.



6. Find the sum of the interior angles of a hexagon.

7. Find the measure of each interior angle of a regular octagon.

8. Find the measure of each exterior angle of a regular 12-gon.

9. A regular polygon has exterior angles of $40°$. How many sides does it have?

10. The interior angles of a polygon sum to $1440°$. How many sides does it have?

Geometry Review: Unit 2 — Angles & Parallel Lines

Answer Key

Station 1: Angle Pairs: Vertical, Linear, Complementary, Supplementary

1. $x = 10;\ 60°$ each
2. $108°,\ 72°$ $5x + 20 = 180,\ x = 32$
3. $43°$
4. $52°$
5. $74°,\ 106°,\ 106°$
6. $35°,\ 55°$
7. $111°,\ 69°$ $6x + 6 = 180,\ x = 29$
8. $x = 20;\ 70°$
9. $30°,\ 60°,\ 90°$
10. $x = 16$

Station 2: Parallel Lines Cut by a Transversal

1. $65°$ *corresponding angles*
2. $112°$ *alternate interior angles*
3. $132°$ *same-side interior angles are supplementary*
4. $125°$ *alternate exterior angles*
5. $x = 10;\ 47°$ *corresponding angles are equal*
6. $x = 15;\ 65°$ *alternate interior angles are equal*
7. $x = 31;\ 77°,\ 103°$ $5x + 25 = 180$
8. $x = 9;\ 59°$ *alternate exterior angles are equal*
9. $\angle 4, \angle 5, \angle 8 = 70°;\ \angle 2, \angle 3, \angle 6, \angle 7 = 110°$
10. congruent; supplementary

Station 3: Proving Lines Are Parallel

1. $x = 12$
2. $x = 16;\ 41°$
3. $x = 14;\ 66°,\ 114°$ $10x + 40 = 180$
4. $x = 11;\ 76°$
5. $x = 40$ $3x + 60 = 180$
6. Converse of the Alternate Interior Angles Theorem
7. They are parallel.
8. Yes; $118 + 62 = 180$, so they are supplementary
9. No; corresponding angles are not congruent
10. Yes; slope of $q = \frac{4}{6} = \frac{2}{3}$

Station 4: Triangle and Polygon Angle Sums

1. $61°$
2. $30°,\ 60°,\ 90°$
3. $55°,\ 65°,\ 60°$ $6x + 30 = 180,\ x = 25$
4. $119°$
5. $x = 30;\ 140°$
6. $720°$ $(6 - 2) \cdot 180$
7. $135°$ $\frac{(8 - 2) \cdot 180}{8}$
8. $30°$ $\frac{360}{12}$
9. 9 $\frac{360}{40}$
10. 10 $(n - 2) \cdot 180 = 1440$